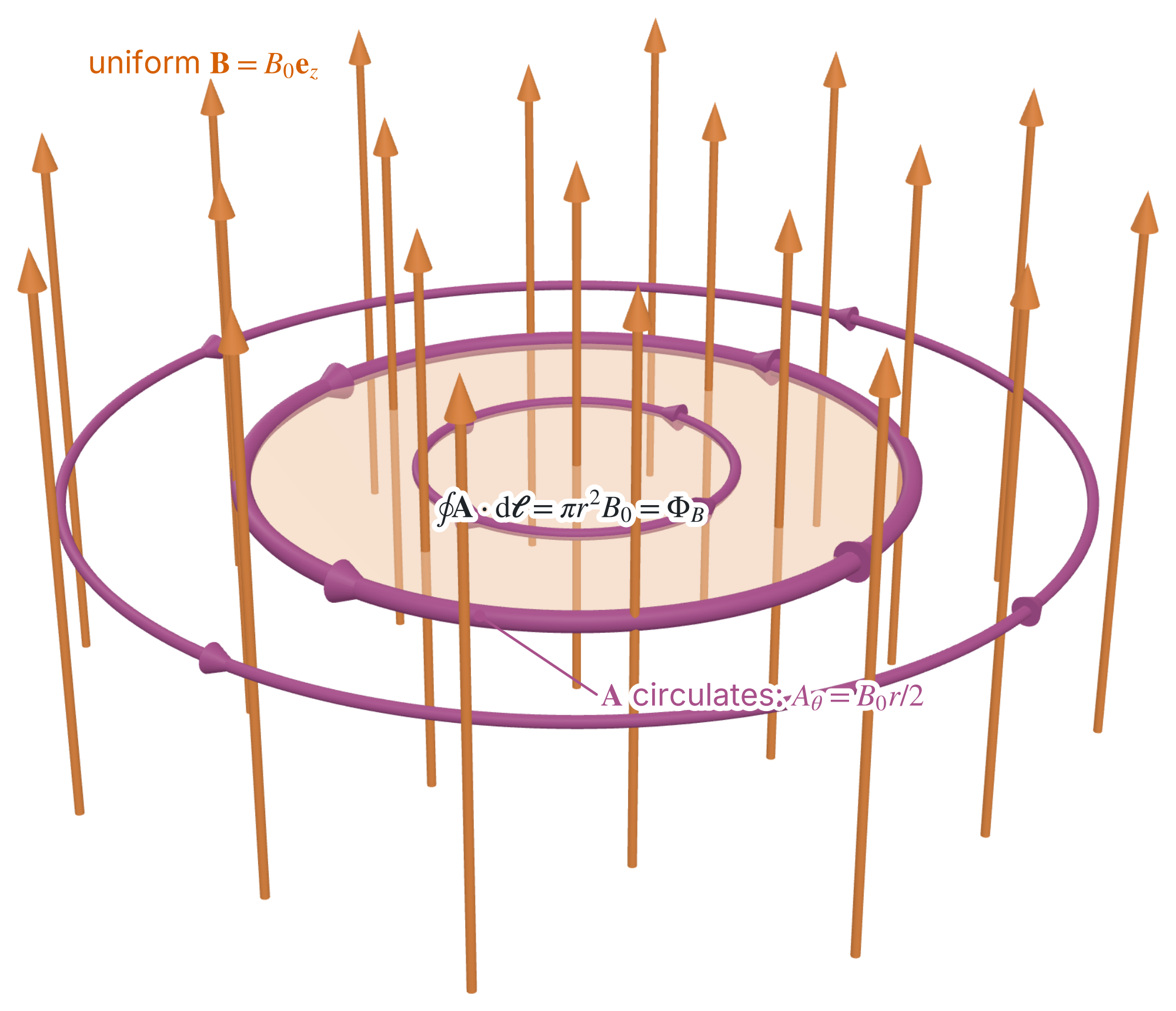

uniform $\mathbf{B} = B_0 \mathbf{e}_z$
$\oint \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} = \pi r^2 B_0 = \Phi_B$
$\mathbf{A}$ circulates: $A_\theta = B_0 r/2$